

CAPSTONE PROJECT REPORT

End-to-End Machine Learning & Computer Vision

Submitted by

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Abstract

AI is associated with handling diverse types of data, from basic Excel sheets to even drone imagery in the modern world. Capstone Project is the ultimate representation of the entire machine learning pipeline process that includes various stages like basic Exploratory Data Analysis (EDA) to computer vision. There are five primary tasks included in the capstone project which are EDA, Supervised Classification, Continuous Regression, Unsupervised Clustering, and Advanced Object Detection.

A really unique part of this assignment was the personalized **SEED** constraint. A dynamic seed was created from a cryptographic hash of my student credentials and enforced algorithmic handicaps such as limiting the depth of the trees and removing important variables. During the project, Logistic Regression and Decision Trees were implemented for predicting the quality of wines, Linear Regression for house prices, and K-means clustering for the Iris flower dataset.

The most interesting thing about this project was the successful implementation and optimization of YOLOv8 Nano architecture on the large-scale VisDrone dataset of 2GB. After training the model for 30 epochs on a cloud GPU, the model became capable of detecting very small and densely packed objects from drone aerial video data, thus satisfying the advanced bonus criteria of the module.

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Chapter 1

Introduction

1.1 Background and Motivation

Machine learning is not only about algorithm invocation but also about tuning it to the specific reality of data that one has. As opposed to projects intended for beginners that provide absolutely clean sets of data, in real engineering one needs to deal with missing features, non-converging models, and significant computational limitations. The key motivation for this project was the simulation of such real-life situations through constructing an automated pipeline of transforming data into predictive models.

1.2 Problem Statement

The core challenge was to design and evaluate multiple AI models across different datasets while strictly following personalized algorithmic rules. Some specific hurdles included:

- Dealing with unscaled features that caused mathematical non-convergence.
- Forcing unsupervised algorithms to group data into non-biological cluster counts.
- Detecting extremely small, overlapping objects (like cars and pedestrians) from a drone's perspective.

1.3 Objectives

1. **EDA & Preprocessing:** Visualize feature correlations using the Mobile Price Classification dataset.
2. **Supervised Learning:** Compare Logistic Regression against a computationally handicapped Decision Tree on the Wine Quality dataset.
3. **Regression:** Predict housing prices while dynamically dropping features based on strict seed constraints.

4. **Unsupervised Learning:** Apply K-Means clustering and visualize 4D data in 2D using Principal Component Analysis (PCA).
5. **Advanced Vision:** Fine-tune a YOLOv8 neural network on the VisDrone dataset, and build a Convolutional Neural Network (CNN) for MNIST digit classification.

1.4 Scope and Limitations

This project was developed using a Google Colab environment powered by a T4 GPU. The biggest limitation was the strict enforcement of my personalized **SEED** (47090249). This seed intentionally crippled some of my models by forcing a low tree depth (`max_depth=6`), removing predictive columns like **HouseAge**, and forcing $k = 4$ on a biological dataset that only has 3 species. Furthermore, YOLOv8 training was capped at 30 epochs and a 640×640 resolution due to runtime limits.

Chapter 2

Technology Overview

To ensure the project was reproducible and utilized GPU acceleration, I relied on the following open-source stack:

- **Environment:** Google Colab (T4 GPU Accelerator)
- **Core Language:** Python 3.12
- **Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Classical Machine Learning:** Scikit-Learn (Linear/Logistic Regression, Decision Trees, K-Means, PCA)
- **Deep Learning & Computer Vision:** TensorFlow/Keras (for the CNN architecture), Ultralytics YOLOv8 (for object detection), and OpenCV (for rendering images).

Chapter 3

Methodology

3.1 System Initialization & Seed Generation

In order to ensure the absolute uniqueness and reproducibility of my results, I used a SHA-256 hash of my name, roll number, and a salt value. This helped me to generate my global SEED: 47090249 and SIG: 02ce8a497ee1, which controlled all random states, the 80/20 train-test splits, and the algorithmic handicaps.

3.2 Task Pipelines

- **Task 1 (Mobile Price EDA):** Loaded the CSV data, checked for nulls, and generated distribution charts and correlation heatmaps. I discovered a very strong positive correlation between a phone's RAM and its price range.
- **Task 2 (Wine Quality Classification):** After dropping the non-predictive Id column, I trained a Logistic Regression model (`max_iter=3000`) and a Decision Tree that was restricted to `max_depth=6`.
- **Task 3 (California Housing Regression):** I had to drop the `HouseAge` column based on the modulo math of my SEED. I then trained a Linear Regression model and plotted the Actual vs. Predicted values.
- **Task 4 (Iris Clustering):** I applied K-Means clustering with $k = 4$ (calculated via $2 + (\text{SEED} \% 3)$). I then used a PCA model to squash the 4-dimensional biological data into a 2D scatter plot so it could be easily visualized.
- **Task 5 (YOLO & CNN):**
 - *Track C (YOLO):* I used the built-in `VisDrone.yaml` script to automate the download of the entire 2GB aerial dataset. I initialized `yolov8n.pt` and fine-tuned it for 30 epochs, rendering bounding boxes on the validation images.

- *Track B (CNN)*: I built a simple sequential 2D Convolutional Neural Network, trained it on the MNIST digits dataset for 3 epochs, and plotted a 3×3 grid to validate the predictions.

Chapter 4

Results and Discussion

4.1 Classification Results (Wine Quality)

- **Logistic Regression:** Accuracy: 65.07%, Precision: 61.57%, F1-Score: 0.6301
- **Decision Tree:** Accuracy: 56.77%, Precision: 55.46%, F1-Score: 0.5565

Discussion: The logistic regression was clearly better than the decision tree classifier because of its greater accuracy. The accuracy of the tree was limited due to the `max_depth=6` parameter and, hence, caused underfitting of the data by the tree. One other thing that I observed in the case of the logistic regression is the warning about non-convergence. This made me understand that the data needed to be normalized for perfect mathematics.

4.2 Regression Results (California Housing)

Metrics: MAE: 0.5359, MSE: 0.5560, RMSE: 0.7456, R^2 : 0.5862.

Discussion: The value of 0.58 for R^2 indicates moderate predictive capabilities. In examining the scatter plot, it is evident that there is an abnormal horizontal cluster of points at the very top. The truth is that the data set was deliberately set to have an upper limit on the price of the houses as \$500,000. Since my model was predicting realistic prices beyond this false upper limit, they were taken to be errors.

4.3 Clustering Results (Iris PCA)

Discussion: K-Means was excellent at creating a perfectly pure cluster for the *Setosa* species. Unfortunately, as my seed required me to choose $k = 4$, there was nothing much I could do to avoid breaking down the overlapping *Versicolor* and *Virginica* species with some extra clustering by force.

4.4 Advanced Vision Results (VisDrone YOLOv8)

Discussion: In 30 training epochs, the model successfully managed to decrease its loss function, giving an mAP@50 score of 0.283. It had no issues in recognizing large objects like Cars (0.711 mAP50) and Buses (0.385 mAP50) but was quite challenged when it came to recognizing tiny objects such as bicycles. It is impressive that the pre-trained nano model performed well despite the high density of the drone data and the low 640×640 resolution.

Chapter 5

Conclusions and Future Scope

5.1 Conclusion

The capstone project is definitely a highly satisfying and comprehensive approach for putting all the aspects of machine learning together into an automated workflow. The output was clear evidence of the importance of feature engineering, data scaling, and tuning of the parameters for the quality of the model itself. With completing the more complicated YOLOv8 VisDrone tracking challenge and the Bonus part, I demonstrated my ability to go from working with tabular data to deploying actual deep learning algorithms in the cloud environment.

5.2 Future Scope

To take these prototype models to a production-ready level, I would make the following improvements:

1. **Enhancing Pipelines:** Incorporate `StandardScaler` in the classification and regression pipelines to eliminate the convergence errors and boost the accuracies.
2. **Flexibility for Algorithms:** Eliminate the arbitrary constraint of `SEED` so that I can optimize my models by using grid search and letting them discover their optimal hyperparameters.
3. **Improving Computer Vision:** Regarding the drone detection, I would increase the image size to `imgsz=1280` and use SAHI. Training for 100+ epochs on good hardware will significantly enhance my model's performance in detecting small occluded pedestrians.